

CTR4A01 MODBUS RTU Command

MODBUS command (function code, write 06/16, read 03/04)

Demo 1 : <https://youtu.be/9wDJ55X0n9Y>

Demo 2 : <https://youtu.be/ovTkTU-KEJA>

Note :

1 MODBUS command must be HEX

2 Slave address (Device address) must be the same as the setting. You can also use this command to query the current device address: FF 03 00 FD 00 01 00 24

3 The Baudrate and parity should be consistent

4 If communication fails, please short the RES jumper for 5 seconds to restore the factory settings

Supported function codes:

Function Code	Modbus Address (PLC)	Register Address	Describe
04:	30001	0x0000 (0)	AC current value register (unit: mA) 5A:0-5000; 10A:0-10000; 20A:0-20000;
03	40001		
		0x0080-0x00FF (128-255)	Read special function registers (baud rate 485 address, etc.)
06	40001		
		0x0080-0x00FF (128-255)	Write a single special function register (baud rate 485 address, etc.)
16(0x10)	40001		
		0x0080-0x00FF (128-255)	Write multiple special function registers (baud rate 485 address, etc.)

All states are mapped into 4xxx range registers. The user can monitor the input and output status of the module by reading or modifying the value of the 4xxx interval register (03 06 16 function code)

Register address	Register contents	Register value	Remarks	R/W
0x00A0 (160)	AC current value	AC current value register (unit: mA) 5A:0-5000; 10A:0-10000; 20A:0-20000;		R
0x00C0 (192)	Current correction value. If the error between the read current and the actual current exceeds 1%, please enter the correct current value in this register. Read data: 0XFFFF			R

Special function Register :				
0X00F6 (246)	AC current value 0x00A0 register is automatically reported	0: Query function (default) 1-255: Automatically report, the unit is second. 1: Report every 1 second 2: Report every 2 seconds 10: Report every 10 seconds Maximum interval of 255 seconds		R/W
0x00F7 (247)	AC frequency	0 : 50HZ (default) 1 : 60HZ		R/W
0x00FB (251)	Factory Reset	Factory Reset: 1 Short the RES jumper for 5 seconds 2 Enter the following command at the current baud rate: FF 06 00 FB 00 00 ED E5		R/W
0x00FC (252)	Command Return Time	Time interval for command return (unit: 40MS) Setting value: 0-25		R/W
0x00FD (253)	RS485 address (Slave ID)	Read address: FF 03 00 FD 00 01 00 24; Set address to 0x02: FF 06 00 FD 00 02 8C 25		R/W
0x00FE (254)	Baud rate	0-255	0:1200 1:2400 2:4800 3:9600 (default) 4:19200 5:38400 6: 57600 7: 115200 Others: Factory reset	R/W
0x00FF (255)	Parity	0-2	0 None Parity 1 Even Parity 2 Odd Parity	R/W

9600 Band ,8 Data bits,None Parity,1 Stop Bit.

MODBUS commands you can use "Modbus Poll" input, as shown below
(CRC check generated automatically)

Modbus Poll - Mbpoll3

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 22 23 101 ?

Mbpoll1

Tx = 3478: Err = 4: ID = 1: F = 03: SR = 1000ms

	Alias	00000
0		
1	output 1	0x0001
2	output 2	0x0001
3	output 3	0x0001
4	output 4	0x0000
5	output 5	0x0001
6	output 6	0x0000
7	output 7	0x0001
8	output 8	0x0001

Mbpoll3

Tx = 8602: Err = 567: ID = 1: F = 03: SR = 1000ms

	Alias	00250
0		
1		
2		
3	Input-output relationship	1
4	Baud rate	3
5		
6		
7		
8		

Mbpoll4

Tx = 4296: Err = 561: ID = 1: F = 03: SR = 1000ms

	Alias	00120	Alias	00130
0			Input 2	0x0000
1			Input 3	0x0000
2			Input 4	0x0000
3			Input 5	0x0000
4			Input 6	0x0000
5			Input 7	0x0000
6			Input 8	0x0000
7				
8				
9	Input 1	0x0000		

You can also use HyperTerminal serial input, as shown below
(Manually add CRC check)

Serial Port Tester

Port Options Help

Ready: COM1

☐ Loops Unit/ms Send

1. Read the AC current value:

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2))
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2))
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Modbus Address(PLC): 30001

RS485 address : 0x01~0x3F

Function code: 0x04

Register address:0x0000

Read number :0x0001

For example, Read the AC current value of channel 0:

Send data(address 1): 01 04 00 00 00 01 31 CA

Return data : 01 04 02 07 D0 BA 9C

01 RS485 address, 04 function code, 02 length, 07D0 represents the value of channel 0, which is 2000 in decimal, Indicates that the AC current value of channel 0 is 2000mA.

In addition, the AC current value is also mapped to the 40000 interval register. The user can read the value of the AC current value through the 03 function code.

Modbus Address(PLC): 40161

RS485 address : 0x01~0x3F

Function code:0x03

Register address:0x00A0

Read number: 0x0003

For example:

Send data(address 1): 01 03 00 A0 00 01 84 28

Return data : 01 03 02 07 D0 BB E8

01 RS485 address, 03 function code, 02 length, 07d0 represents the value of channel 0, which is 2000 in decimal, that the AC current value of channel 0 is 2000mA..

2. Set the Current correction value (correction value) :

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
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Modbus Address(PLC): **40193**

RS485 address : 0x01~0x3F

Function code:Write 0x06/0x10,Read 0x03

Register address:0x00C0

Current correction value. If the error between the read current and the actual current exceeds 1%, please enter the correct current value in this register. Read data: 0xFFFF

For example : The actual current is 1000MA, but the read current is 1200MA, and the deviation is greater than 1%. Then enter 1000 in this register to calibrate.

Send frame: **01 06 00 C0 03 E8 89 48**

Return frame: **01 06 00 C0 03 E8 89 48**

The return frame is the same as the send frame. 03 E8 means correction current is 1000

Special function Register

1.Set the RS485 address(Slave ID)

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
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Modbus Address(PLC): 40254

RS485 address :0x01~0Xf8/0XFF

Function code:Write 0x06/0x10,Read 0x03

Register address:0x00FD(253)

Value: 2 bytes (values 1-248)

For example 1: Set the current device address to 0x02

Send data(address is 1): 01 06 00 FD 00 02 99 FB

Return data : 01 06 00 FD 00 02 99 FB

Send data(don't know the address): FF 06 00 FD 00 02 8C 25

Return data : FF 06 00 FD 00 02 8C 25

For example 2: Read device address(0X0001)

Send data : FF 03 00 FD 00 01 00 24

Return data : 01 03 02 00 01 79 84

Note: With this command, there can be only one module on the bus 485,
More than one will go wrong!

2.Write baud rate

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
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Modbus Address(PLC): 40255
RS485 address :0x01~0x3F
Function code:Write 0x06/0x16;Read 0x03
Register address:0x00FE(254)
Value: 2 bytes (values 0-7)

For example 1, Change the baud rate to 4800bps:
Send data(address 1):01 06 00 FE 00 02 69 FB
Return data :01 06 00 FE 00 02 69 FB

Baud rate corresponds to the number: 0:1200 1:2400 2:4800 3:9600
4:19200 5:38400 6:57600 7: 115200 8: Factory reset

Note: 1 The baud rate will be updated only when the module is powered on again when this command is used!

2 When the number corresponding to the baud rate is 8, the factory settings can be restored

For example:01 06 00 FE 00 08 E9 FC

For example 2 Read the current baud rate:
Send data(address 1):01 03 00 FE 00 01 E5 FA
Return data :01 03 02 00 03 F8 45

01 RS485 address, 03 Function, 02 length, F8 45 crc16, 03 means the current baud rate is 9600bps

Baud rate corresponds to the number: 0:1200 1:2400 2:4800 3:9600
4:19200 5: 38400 6:57600 7: 115200

5. Set Command(Date) Return Time

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2))
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2))
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Modbus Address(PLC): 40253
RS485 address :0x01~0x3F
Function code:Write 0x06/0x16;Read 0x03
Register address:0x00FC(252)

Value: 2 bytes (values 0-25)

For example, set the data return delay to 200ms

Send data(address 1):01 06 00 FC 00 05 89 F9

Return data :01 06 00 FC 00 05 89 F9

Return the delay time calculation formula: $X = 05 * 40 = 200\text{MS}$

Note: The maximum can be set to 1000MS. If it exceeds 1000MS, that is, the setting value is greater than 25, and the data return delay will be initialized.

That is: 01 06 00 FC 00 20 48 22 can make the data return delay to restore initialization 0

6. Set Parity

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
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Modbus Address(PLC): 40256

RS485 address :0x01~0x3F

Function code:Write 0x06/0x16;Read 0x03

Register address:0x00FF(255)

Value: 2 bytes (values 0-2)

For example, set the parity to Even parity

Send data(address 1):01 06 00 FF 00 01 78 3A

Return data :01 06 00 FF 00 01 78 3A

0 None Parity 1 Even Parity 2 Odd Parity

Note: 1. When using this command, the module is powered on again, and the check digit will be updated!

2. When the setting is greater than 2, the default value will be restored to 0 after powering on again, and there will be no verification.

7. Factory reset:

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16 (2)
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Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16 (2)
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Modbus Address (PLC): 40252

RS485 address : 0x01~0x3F

Function code: Write 0x06;

Register address: 0x00FB (251)

Send data (address 1): FF 06 00 FB 00 00 ED E5

Return data : FF 06 00 FB 00 00 ED E5

It can also be reset by hardware: short the RES jumper of the board for 5 seconds, and then power on again.